

SSD21, Educational Toolbox for Static, Stability, and Dynamic Analysis of Frame

Truong Thanh Chung

Faculty of Civil
Engineering

Nha Trang University
Nha Trang, Vietnam
chungtt@ntu.edu.vn

Le Nguyen Anh Vu

Faculty of Civil
Engineering

Nha Trang University
Nha Trang, Vietnam
vulna@ntu.edu.vn

Le Cong Lap

Faculty of Civil
Engineering

Nha Trang University
Nha Trang, Vietnam
laplc@ntu.edu.vn

Nguyen Thang Xiem

Faculty of Civil
Engineering

Nha Trang University
Nha Trang, Vietnam
xiemnt@ntu.edu.vn

Abstract: Nowadays, commercial structural analysis software is both powerful and easy to use. That combination comes with the danger of being a “black box” where users provide the input and take the output from the software without understanding of how it works. On the other hand, hand-calculation methods currently taught in university such as force method and displacement method are completely out-of-date since they only allow solving relatively simple problems. The authors have developed an open-source MATLAB-based educational structural analysis program to use as a tool for undergraduate and graduate classes in order to encourage students to understand the underlying principles of Finite Element Method and apply for static, stability, and dynamic analysis of frame structures.

Keywords: educational toolbox, static analysis, stability analysis, dynamic analysis, finite element method

I. INTRODUCTION

Structural analysis due to the effects of static and dynamic loads is an essential part of civil engineering programs. The modern approach to teach structural analysis is using programming tools where students learn the computational methods, such as the finite element method to analyze internal forces of structures subjected to static loads, or direct integration method to analyze vibration of structures subjected to dynamic loads. Several programs on MATLAB platform were developed for this purpose, such as CALFEM [1], NUMA-TF [2], FEM Toolbox for Solid Mechanics [3], MASTAN2 [4], VisualFEA [5]. The authors have looked closely into each of those programs, unfortunately we found that they all have serious drawbacks: CALFEM is lack of capability to model frame structures with internal hinges; NUMA-TF is coded using Object Oriented Programming (OOP) method, which is extremely difficult for students to understand; FEM Toolbox for Solid Mechanics has a bug that caused it to produce the wrong internal forces diagrams; and finally, both MASTAN2 and VisualFEA are closed-source code. For that reason, the authors have developed an open-source educational structural analysis toolbox (SSD21) based on MATLAB platform [6]. Much effort was made to simplify the code so that it can be understood and easy to use by students in classes. The source code is completely free to download, use, and modify in the following address: <https://github.com/chungntu/SSD21>. This paper is served as a short introduction to SSD21. Students are highly recommended to download and review the provided source code for better understanding.

Several improvements compared to other programs as mentioned above are as follows. First, the results of the toolbox SSD21, such as node displacement, internal forces diagrams, modal analysis, stability analysis are shown to be very close to professional commercial software, such as SAP2000 [7]. Second, we introduced the ability to model internal hinges in frame structures, which is important for a wide class of structural analysis. Third, we introduced many options in the code to allow users to change the appearances of the outputs: for example, show/hide node and beam numbering, show/hide numerical value and numerical scale bar of internal force diagrams. Finally, a non-linear structural analysis (P-delta effect) was developed for advanced users. The flowchart of the FEM program SSD21 for static and dynamic analysis is shown in Fig. 1.

In the next section, several examples of structural analysis are shown to illustrate the capabilities of SSD21 toolbox, including static analysis of beam structure, static and stability analysis of concentrically braced frame, and modal analysis of frame structure. It is noted that for the purpose of illustration, the information regarding numerical values of dimensions and load amplitudes was neglected for concision. The final section is the conclusion.

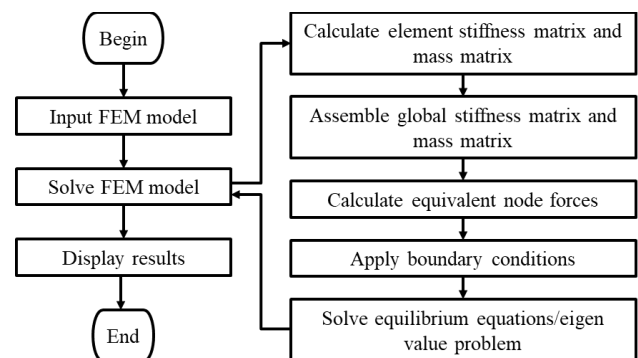


Figure 1. Flowchart of FEM program SSD21 for static and dynamic analysis

II. EXAMPLES

A. Static analysis of beam structure

In this problem, the beam structure is subjected to uniform force, concentrated force, and concentrated moment as shown in Fig. 2a. Both ends of the middle beam (draw using red color) are hinges. The uniform force in vertical direction is illustrated as a dot line, with the dot line above the beam meaning that the force direction is pointed down. The



concentrated force is illustrated as an arrow, and the concentrated moment is illustrated as an arc-arrow.

The deformed shape of the beam, the shear forces diagram, and the moment diagram are shown in Fig. 2b, Fig. 2c, and Fig. 2d, respectively. These figures are exported automatically as both MATLAB files (.fig) and picture files (.jpeg) and saved in the same directory of the input text file (.s2k). Fig. 3a, b show the shear forces and the moment diagrams from SAP2000® for comparison. Both figures are numerically identical with corresponding diagrams from SSD21 (Fig. 2c, d), except that the shear force diagram by SAP200 is upside down. Table 1 shows the node displacement of the beam structure.

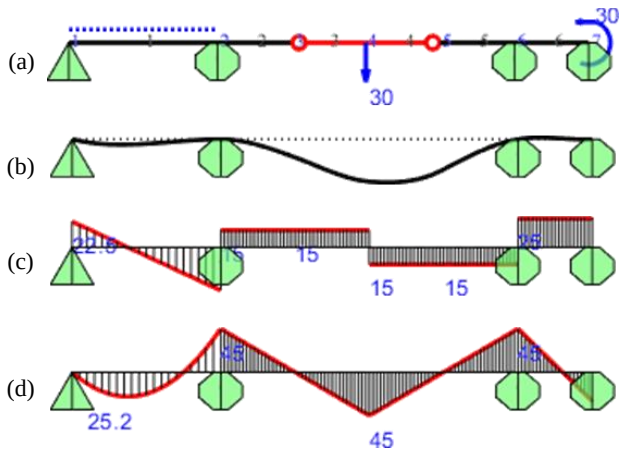


Figure 2. Top to bottom. (a) Beam structure subjected to static loads, (b) Deformed shape, (c) Shear forces diagram, (d) Moment diagram

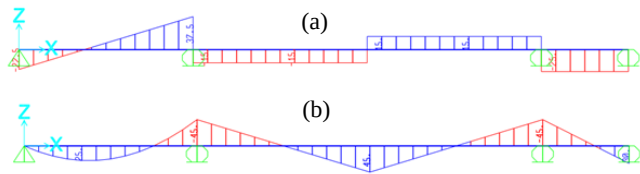


Figure 3. Top to bottom. (a) Shear forces diagram, (b) Moment diagram of the beam using SAP2000

Table 1. Node Displacement of Beam Structure Example A

Node #	X-direction [m]	Z- direction [m]	Rotation [rad]
1	0.00000000	0.00000000	-0.00000023
2	0.00000000	0.00000000	0.00000000
3	0.00000000	-0.00000068	-0.00000034
4	0.00000000	-0.00000158	-0.00000008
5	0.00000000	-0.00000113	0.00000049
6	0.00000000	0.00000000	0.00000015
7	0.00000000	0.00000000	0.00000004

B. Static analysis of concentrically braced frame

A concentrically braced frame subjected to static loads is shown in Fig. 4a. The frame elements in the middle bay have

hinges at both ends, therefore they are effectively act as truss elements. The loads acting on the structures are uniform forces on each floor, positive wind loads at the left side of the frame, and negative wind loads at the right side of the frame. For the uniform load in vertical direction, a line of triangle is used for illustration.

The deformed shape is shown in Fig. 4b. The normal forces diagram, shear forces diagram, and moment diagram are shown in Fig. 4c, Fig. 4d, and Fig. 4e, respectively. Table 2 shows the node displacement of the frame structure. It is noted that for complex structures such as in this example, it is recommended to use a scale line to show the magnitude of the displacement and forces rather than showing directly as in Example A.

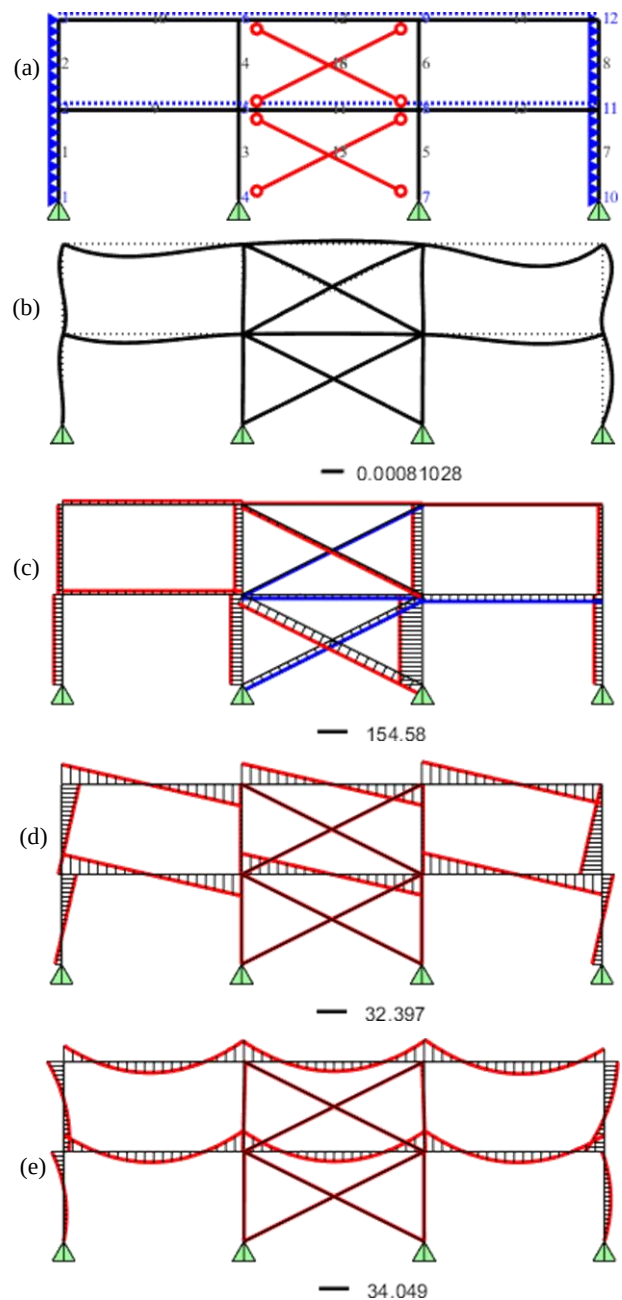


Figure 4. Top to bottom. (a) Concentrically braced frame subject to static loads, (b) Deformed shape, (c) Normal forces diagram, (d) Shear forces diagram, (e) Moment diagram

Table 2. Node Displacement of Concentrically Braced Frame in Example B

Node #	X-direction [m]	Z- direction [m]	Rotation [rad]
1	0.00000000	0.00000000	-0.00012326
2	0.00002691	-0.00000864	-0.00034288
3	0.00004337	-0.00001294	-0.00046463
4	0.00000000	0.00000000	-0.00001778
5	0.00001930	-0.00001195	0.00001625
6	0.00003568	-0.00001987	0.00006220
7	0.00000000	0.00000000	-0.00000373
9	0.00002700	-0.00002319	-0.00001955
10	0.00003338	-0.00003319	-0.00010830
11	0.00000000	0.00000000	-0.00049051
12	0.00004117	-0.00000846	0.00037734

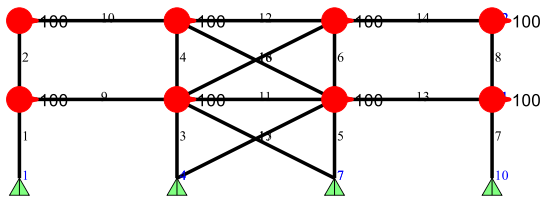
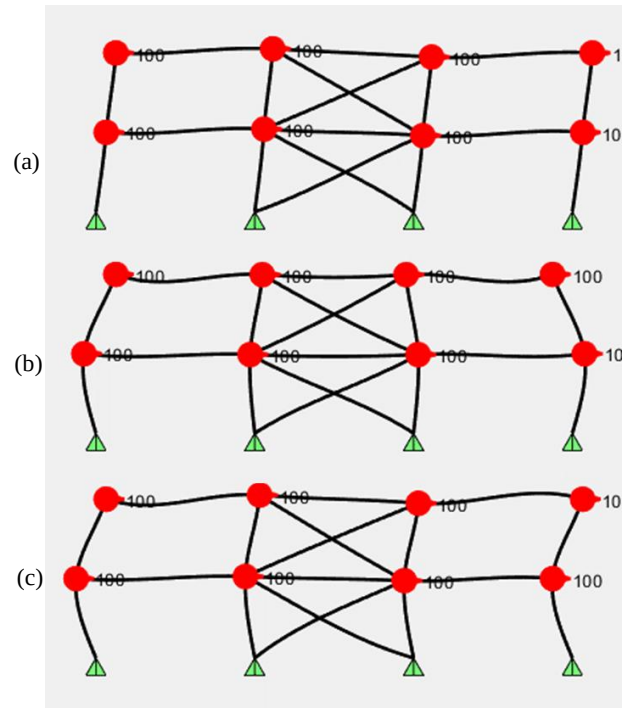
C. Modal analysis of frame structure

In this example, the modal analysis of a plane frame is carried out (Fig. 5). At each node of the frame, there is a concentrated mass acting in horizontal direction, which is illustrated by a horizontal arrow. The self-weight of the structure is also considered in the modal analysis.

As the result of the modal analysis, Fig. 6a, Fig. 6b, and Fig. 6c show the modal mode shape 1, modal mode shape 2, and modal mode shape 3 of the frame structure, respectively. Table 3 shows the solutions of circular natural frequencies ω (rad/s) of the frame structure using SSD21 toolbox compared to SAP2000, which are nearly identical. Though not shown here, the mode shapes are able to animated and exportable as move files (.avi), which is very useful for students to visualize how the system behaves in each mode.

Table 3. Circular Natural Frequencies of The Frame Structure with Comparison to SAP2000

Mode #	Circular Natural Frequencies ω (rad/s)		
	SSD21	SAP2000	Differences (%)
1	56.571488	56.571476	2.1E-05
2	132.999642	132.999571	5.3E-05
3	144.036821	144.036770	3.5E-05
4	157.984938	157.984930	5.1E-06
5	264.952482	264.952481	3.8E-07
6	322.450038	322.450013	7.8E-06

**Figure 5.** Concentrated mass structure**Figure 6.** Top to bottom. (a) Modal mode 1, (b) Modal mode 2, (c) Modal mode 3

D. Stability analysis of concentrically braced frame

In this example, the linear stability analysis (or linear buckling analysis) of a concentrically braced frame is carried out. Fig. 7a shows the frame under the static loads, which is the same as the frame in Example B. The result of buckling analysis is the critical load ratio, which is the parameter that multiplies to the current load that makes the structure buckle.

Fig. 7b, Fig. 7c, and Fig. 7d show the buckling mode 1 to 3 of the frame structure. It is noted that the results presented here are linear analysis, which does not consider the P-delta effect of the structure. We have developed an advanced version of stability analysis based on non-linear algorithm; readers who are interested may refer to the source code [8] for further information.

Table 4 compared the critical load ratios of the SSD21 compared SAP2000, which shows a difference of less than 2%. The buckling mode shapes are also able to be animated and exportable as move files (.avi).

Table 4. Linear Bulking Analysis Critical Load Ratio Compared to SAP2000

Buckling Mode #	Critical Load Ratio		
	SSD21	SAP2000	Differences (%)
1	336.238797	340.723475	1.31
2	494.847193	500.897242	1.21
3	646.964575	654.695405	1.18
4	869.997952	881.29009	1.28
5	931.286847	931.560272	0.03
6	1246.659921	1266.497852	1.57

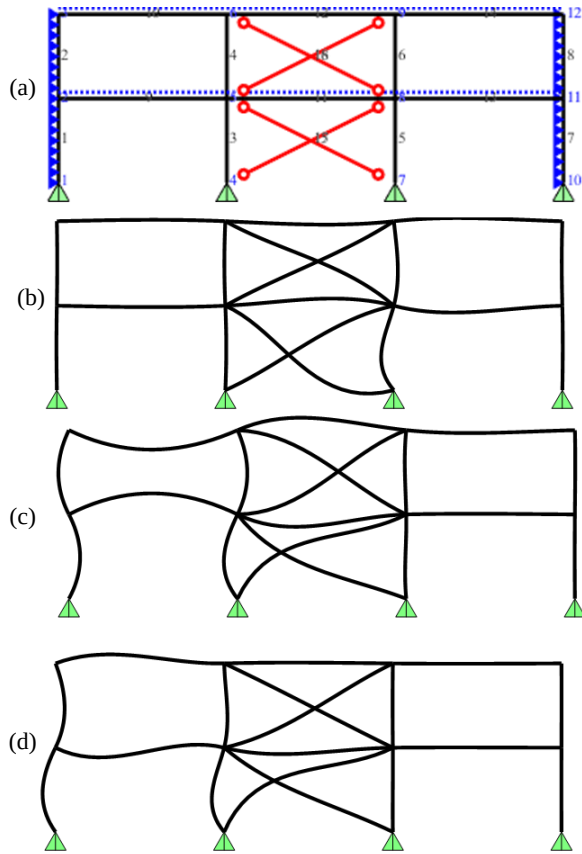


Figure 7. (a) Structure subject to static loads, (b) Bulking mode 1, (c) Bulking mode 2, (d) Bulking mode 3

III. CONCLUSION

In this paper, a review of the educational toolbox for static, stability, and dynamic analysis of frame structure (SSD21) is

presented. We have been using SSD21 in actual teaching at Nha Trang university and it has shown its effectiveness as a teaching and learning tool. New ideas are continuously being added to SSD21 so that the toolbox gets more convenient to use for both students and lecturers.

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